ACoP 2026 Hackathon brief

Beyond CTG Repeats: Advancing Patient Stratification in Myotonic Dystrophy Type 1 (DM1) | Team 1 working summary of the kickoff session, 11 October 2026

The question: Can machine learning identify additional predictors and improve patient stratification beyond traditional covariate modelling?
The task: for each of three treatment arms, find which baseline characteristics drive response, separate real predictors from planted noise, compare covariate models with ML models, and explain the findings clinically.

1. Clinical background

The disease as one chain

1. A stutter in one gene. The *DMPK* gene contains the letters CTG repeated in a row. Normal is up to about 37 repeats; DM1 is 50 to several thousand.
2. Sticky RNA. The cell's RNA copy of the long repeat folds into a clump that stays in the nucleus.
3. A trapped editor. The clump traps a protein (MBNL) that normally edits other RNA messages. This editing is called splicing.
4. Faulty proteins. With the editor trapped, many proteins are made in the wrong version.
5. Many organs affected. One genetic fault, many different problems: a multisystem disease.

What goes wrong

System	Problem
Muscle	**Myotonia** (slow to relax after gripping) and **weakness** (loss of force). These are separate problems
Heart	Slow or irregular electrical conduction
Brain	Poor concentration, daytime sleepiness
Metabolism	Insulin resistance
Eye	Early cataracts
Breathing, gut	Weak breathing, swallowing trouble

Why patients differ

- **More repeats, earlier and worse disease**, in general: 50 to 100 mild or late onset; up to 1,000 classic adult; above that, childhood or congenital.
- **The count is a rough guide.** Repeats keep growing during life at different rates in different tissues (somatic mosaicism).
- **Inherited from one parent.** The repeat often grows when passed on, especially from the mother.
- **The organ mix varies** between patients with the same count.

The three drugs and their outcomes in the dataset

Arm	Where it acts	Outcome	Better is	Organizers' hint on predictors
Mexiletine	The symptom: blocks sodium channels to calm over-excitable muscle	Change in myotonia severity, week 12	Lower (negative)	Clinical presentation may matter
Oligonucleotide	The root: binds the toxic RNA so it is destroyed, freeing the editor protein	Change in muscle strength score (QMT), week 26	Higher	Molecular and clinical burden may both matter; not every disease marker is informative
Tideglusib	Mid-chain: blocks an overactive enzyme (GSK3-beta) in several tissues	Combined cognitive + motor score, 0 to 20, week 26	Higher	Features from more than one domain may contribute

Each arm also has a responder / non-responder label derived from its outcome. Real-world caution from the session: mexiletine relieves stiffness but did not improve walking distance in a six-month trial, so symptom relief is not the same as functional gain.

What the 18 baseline columns represent

Group	Columns	Meaning
Genetic burden	`ctg_repeats`, `somatic_mosaicism_index`, `family_history`	Size of the fault, how unstable it is, which parent it came from
Disease timeline	`age`, `age_at_onset`, `disease_duration`, `phenotype`	How early it started and how long it has run; `phenotype` = juvenile, adult or late onset
Muscle	`myotonia_severity`, `muscle_strength`, `ck_level`	Stiffness, force, and a blood marker of muscle damage
Heart / brain	`pr_interval`, `cognitive_function`	Electrical conduction time on ECG; thinking ability
Metabolic, eye, general	`insulin_resistance`, `bmi`, `albumin`, `cataracts`, `sex`, `smoking_status`	Sugar handling, body size, nutrition, lens clouding, demographics, lifestyle

Several columns overlap by nature (age, onset and duration; repeat count, onset and phenotype). Methods may pick one of a group and drop the others, so interpret them as a group. **Some columns are deliberate noise:** "some look plausible but are noise".

2. Approach: suggested analysis steps

Decisions to lock first

Decision	Choice and reason
Unit	Three separate analyses, one per arm. Outcomes are arm-specific
Data shape	500 patients, one row each, about 167 per arm. No doses, concentrations or repeat visits, so no PK or exposure-response model
Outcome	Continuous as primary (the label is derived from it); responder label as secondary. State which and why
Split	Per arm, 75/25, stratified by responder label, fixed seed. Test set locked
Model size	Simple and penalised: small arms

Risks the organizers warned about

- **Leakage:** any cleaning rule, scaling or tuning that uses the test set inflates the scores.
- **Small test sets:** a few dozen patients per arm; one patient moves the numbers. Do not rank methods on small gaps.
- **Correlated covariates:** LASSO may drop a meaningful one because a correlated one took its place.
- **Accuracy alone** misleads when responders are the minority.
- **Importance is not causation:** SHAP explains the model, not biology.

Steps

#	Step	What to do
1	Explore and clean	Check ranges, skew, outliers, label spellings, impossible values and correlated covariates. The dataset has no missing values but needs cleaning. Decide cleaning rules from baseline columns only, never from the outcome, and record each rule in the code
2	Split	Hold out the test set per arm before any modelling
3	Preprocess	Encode categories; learn scaling from the training set only, then apply to the test set
4	Covariate models	Regression baseline, stepwise selection, LASSO, elastic net. Add splines or a GAM only where exploration shows a curved relationship
5	ML models	Random forest; XGBoost untuned and tuned on a small grid by cross-validation inside the training set
6	Evaluate once	One scoreboard per arm, every method on the same test patients. Continuous: RMSE, MAE, R^2. Label: ROC-AUC, recall, precision (accuracy only alongside them; state the cut-off). Add bootstrap intervals
7	Compare covariates	A grid of which method picked which covariate. Agreement builds confidence; a covariate picked by one method needs a second look
8	Signal versus noise	Cross-method agreement, stability under resampling, permutation importance or SHAP, and a random "probe" column as a negative control
9	Interpret	Tie each selected covariate to the drug's mechanism: symptom relief, RNA-level correction, multi-tissue action. Flag any that contradict it
10	Propose stratification	For each arm, say which patients to enrol or stratify on and what that means for trial design. Note the limit: no placebo arm, so predictors show who does well on a drug, not that the drug caused it

Slides: five for a five-minute talk

1. **Answer first:** the hackathon question answered in one sentence.
2. **Workflow:** the six kickoff steps, the split, how noise was handled.
3. **Predictors per arm:** the covariate grid, and which plausible covariates turned out to be noise.
4. **Clinical sense** per arm, with the covariate-versus-ML scoreboard.
5. **So what:** stratification proposal, limits, AI disclosure.

3. Logistics

Dates

When	What
Sun 11 to Tue 13 Oct	Teams work independently. Magnolia 1 is available on 12 and 13 Oct
Tue 13 Oct, 8:00 to 9:00 AM	Office hour, Magnolia 1: non-technical questions and resource guidance. Questions can also be uploaded to the team folder at any time
Tue 13 Oct, 11:59 PM ET	**Slides and all code files uploaded to the Team 1 SharePoint folder.** One representative emails the chairs that the slides are final
Wed 14 Oct, 9:00 to 10:30 AM	Presentations: 5-minute talk + 3-minute Q&A (one judge question). Judges score live. At least one member presents
Wed 14 Oct, 4:30 to 5:00 PM	Winners announced at the closing session

Scoring: 30 points

Category	Judges look for
Clinical (10)	Key predictors per arm; signal separated from noise; interpretation and a stratification proposal
Technical (10)	Sound workflow; unbiased split; justified methods; appropriate metrics
Communication (10)	Clear story in time; supportive visuals; confident Q&A; meaningful conclusion

> "Meaningful covariate identification matters more than fancy models."

Rules and team

- Any platform or language. AI tools are allowed: report which tools and for what purpose, in the code and slides. The team must understand every step and is accountable for results.
- Co-chairs and facilitator support teams but do not judge.
- **Team 1:** Euibeom Shin, Luong Vuong, Vijay Ivaturi, Ming Gao, Wenzhe Ma, Bei Zhao.
- **Roles to fill:** analysis and code; clinical sense-check; slides; presenter and team representative.

Sources: kickoff slide deck and session notes, 11 October 2026. The dataset is simulated and does not describe real patients. Section 2 is a suggested approach for the team to agree, not an instruction from the organizers. Prepared with Claude (Anthropic).